

SHAHRUD, Iran, Islamic Republic of

WMO: 407390

Lat: 36.3801N

Lon: 54.9292E

Elev: 1349

StdP: 86.13

Time Zone: 3.50 (IRN)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.8	-4.1	-15.4	1.2	1.9	-13.2	1.4	2.1	8.1	2.3	6.6	4.5	0.9	330	0.432

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	36.0	17.7	34.6	17.3	33.2	16.8	19.6	31.8	18.8	31.0	18.2	30.1	3.1	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.2	12.7	26.4	14.3	12.0	25.8	13.5	11.4	25.5	62.1	31.8	59.4	31.0	57.2	30.2	24.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.1	7.5	6.4	DB	-7.9	38.7	2.5	1.0	-9.8	39.4	-11.2	40.0	-12.7	40.5	-14.5	41.2	
			WB	-9.2	21.0	2.3	1.1	-10.9	21.8	-12.2	22.5	-13.6	23.1	-15.2	24.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.5	2.3	4.6	9.6	15.2	20.9	25.5	27.8	26.7	22.9	16.8	8.9	3.6
	DBStd	9.64	3.13	3.74	4.41	4.17	3.17	2.99	2.78	2.44	2.91	3.53	4.02	2.98
	HDD10.0	727	239	154	61	8	0	0	0	0	0	1	66	198
	HDD18.3	2088	497	385	273	108	12	0	0	0	3	72	284	456
	CDD10.0	2720	0	3	48	166	338	466	551	518	386	213	32	1
	CDD18.3	1041	0	0	2	15	92	217	292	259	139	26	0	0
	CDH23.3	10335	0	0	11	115	721	2204	3318	2669	1133	165	0	0
	CDH26.7	4486	0	0	1	19	207	979	1643	1218	396	23	0	0
Wind	WSAvg	2.2	1.2	1.8	2.1	2.4	2.7	3.3	3.6	3.2	2.3	1.7	1.4	1.1
Precipitation	PrecAvg	158	15	22	27	26	17	7	4	3	5	6	10	15
	PrecMax	212	46	92	68	91	54	24	20	12	28	29	32	39
	PrecMin	65	0	1	1	2	0	0	0	0	0	0	0	0
	PrecStd	41	11	19	16	20	14	6	6	3	6	7	9	10
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.6	17.0	24.5	29.0	32.5	36.2	38.1	37.0	33.9	29.0	21.5	14.3
		MCWB	5.4	7.2	10.8	13.0	15.2	17.1	18.4	17.2	15.7	15.1	10.8	6.7
	2%	DB	10.3	13.9	20.9	26.6	30.6	34.7	36.5	35.2	32.1	26.8	19.1	11.6
		MCWB	4.4	5.8	9.0	12.5	14.7	16.6	18.2	17.3	15.6	13.5	9.6	5.5
	5%	DB	8.8	12.1	18.7	24.4	29.0	33.2	35.1	33.7	30.6	25.2	17.1	10.0
		MCWB	3.5	5.0	8.3	11.6	14.4	16.2	18.0	17.1	15.2	12.8	8.7	4.7
	10%	DB	7.2	10.6	16.7	22.4	27.4	31.7	33.5	32.3	29.1	23.4	15.2	8.6
		MCWB	2.7	4.3	7.5	11.0	14.0	15.9	17.5	16.7	15.0	12.1	7.8	4.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.6	8.5	11.7	14.2	17.1	19.6	21.2	20.2	18.6	16.3	11.7	7.5
		MCDB	10.7	13.7	21.1	25.0	27.8	32.2	33.2	31.6	29.5	26.8	19.4	12.3
	2%	WB	5.1	6.9	10.3	13.2	15.9	18.4	19.9	19.2	17.7	14.7	10.4	6.3
		MCDB	8.7	12.2	18.2	23.3	27.3	30.9	32.5	30.6	28.4	24.4	17.3	10.5
	5%	WB	4.0	5.7	9.2	12.5	15.2	17.5	19.1	18.6	16.8	13.6	9.4	5.4
		MCDB	8.2	11.0	17.0	21.9	26.5	29.9	31.8	30.0	27.0	22.9	15.9	9.2
	10%	WB	3.1	4.8	8.2	11.7	14.5	16.7	18.5	18.0	16.1	12.7	8.3	4.5
		MCDB	6.9	9.9	15.7	20.7	25.6	29.0	30.9	29.3	26.1	21.7	14.4	7.9
Mean Daily Temperature Range	5% DB	MDBR	7.7	8.8	10.0	10.4	10.6	11.0	10.3	10.7	11.3	10.8	9.0	7.5
		MCDBR	9.2	10.4	11.8	11.9	12.0	12.0	11.5	11.9	12.4	12.2	10.5	9.3
		MCWBR	5.3	5.3	5.1	4.5	4.2	4.0	3.8	4.0	4.6	5.0	4.9	5.3
	5% WB	MCDBR	8.5	9.3	10.3	10.7	11.1	11.1	10.8	10.8	11.1	11.1	9.5	8.1
		MCWBR	5.1	5.2	5.2	4.6	4.2	4.4	4.0	4.1	4.5	4.9	4.7	4.9
Clear-Sky Solar Irradiance	taub	0.315	0.351	0.412	0.469	0.484	0.476	0.500	0.451	0.417	0.404	0.349	0.312	
	taud	2.319	2.203	2.019	1.915	1.913	1.955	1.909	2.041	2.107	2.111	2.248	2.340	
	Ebn at Noon	871	875	846	817	810	814	791	824	831	801	816	847	
	Edh at Noon	96	122	161	189	193	185	192	165	146	133	103	89	
All-Sky Solar Radiation	RadAvg	2.60	3.42	4.58	5.64	6.72	7.75	7.51	7.01	5.80	4.33	3.02	2.32	
	RadStd	0.16	0.29	0.30	0.44	0.26	0.28	0.26	0.36	0.19	0.18	0.18	0.23	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	+0.36	N/A	N/A	+0.89	+0.53	+0.58	N/A	N/A	N/A

Nomenclature: See separate page